

Razão de Segmentos

Divisão Interna de Segmentos



Exemplo $AB = 120 \text{ cm}$
 $k = \frac{3}{2}$

Razão de Segmentos é
 aditiva assim $3 + 2 = 5$
 São 5 partes $\rightarrow 120 \div 5 = 24$
 $3 \cdot 24 = \overline{AP}$ $2 \cdot 24 = \overline{PB}$

Dizemos que

P divide $\overline{AB}$

INTERNA na

razão $k > 0$

if and only if

$P \in \overline{AB} \setminus \{A, B\}$

$$\frac{AP}{PB} = k$$

Obs.: $k = 1$

$k > 1$

$0 < k < 1$

aproximar de A

$k \rightarrow 0$

aproximar de B

$k \rightarrow \infty$

Divisão Externa de Segmentos

Dizemos que Q divide $\overline{AB}$

externamente na razão k:1

Se e somente se

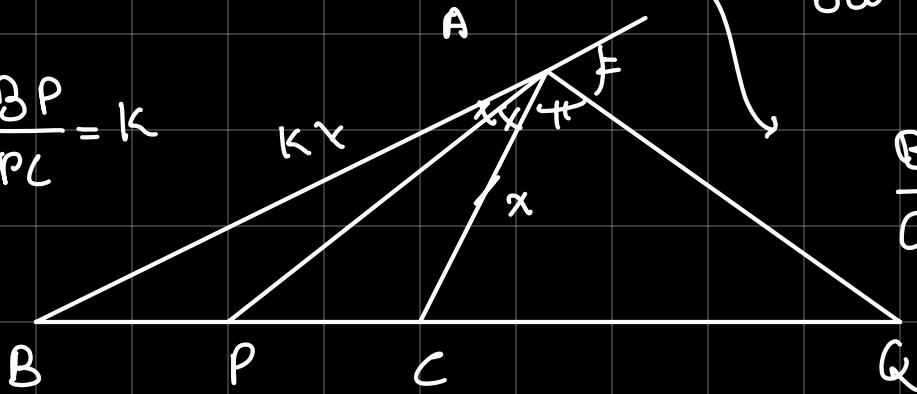
$$Q \in \overleftrightarrow{AB} - \overline{AB}$$

$$\frac{AQ}{BQ} = k$$

é um conceito a posteriori

no problema das Bissetrizes externa

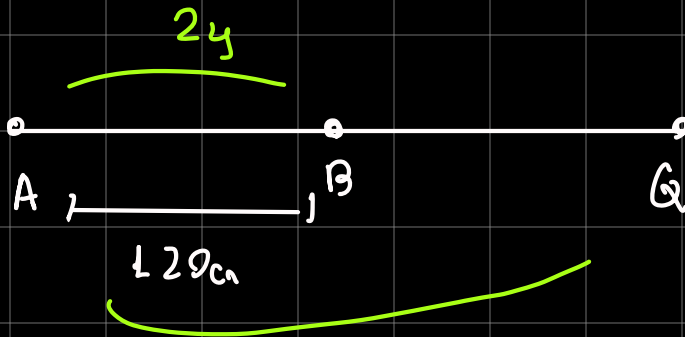
$$\frac{BP}{PC} = k$$



$$\frac{BQ}{CQ} = k$$

Exemplo $AB = 120 \text{ cm}$ $k = \frac{3}{2}$

Divisão
externa
é subtrativa



1 parte é 120 cm

$$AQ = 360 \text{ cm}$$

$$BQ = 240 \text{ cm}$$

$$3y - 2y = y = 120$$

$$3y$$

OBS.:



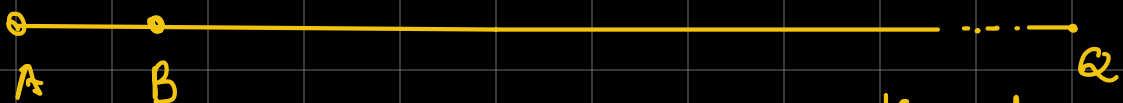
$$k > 1$$

aproximar Q de B
 $k \rightarrow \infty$



$$0 < k < 1$$

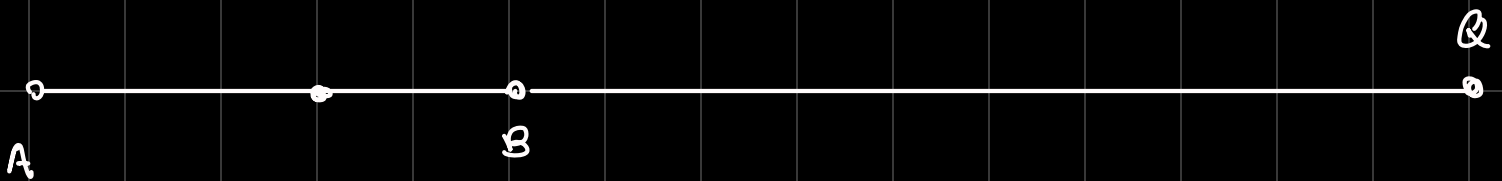
aproximar Q de A
 $k \rightarrow 0$



$$k \rightarrow 1$$

Divisão Harmônica

Dados $\overline{AB}$, $P \in \overline{AB}$, $Q \in \overleftrightarrow{AB} - \overline{AB}$ dizermos que P e Q são Divisores harmônicos de $\overline{AB}$ na razão $k > 0$; se e somente se $\frac{AP}{PB} = \frac{AQ}{QB} = k$



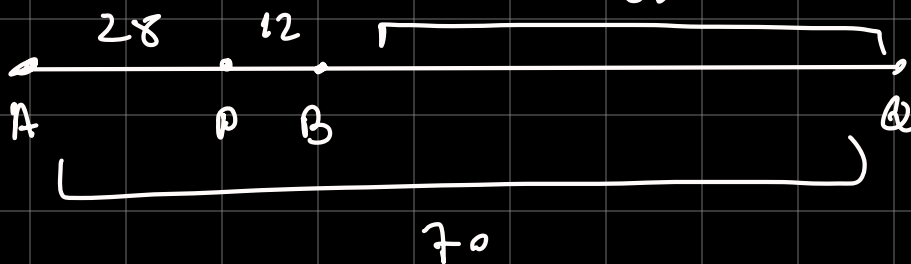
Exemplo $AB = 40 \text{ cm}$ $k = \frac{7}{3}$

$$\frac{AP}{BP} = \frac{AQ}{QB} = \frac{7}{3} \quad \text{construam}$$

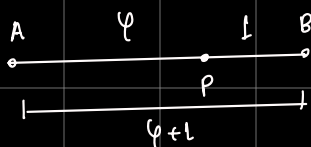
$$u(P) = 7$$

$$u(Q) = 10$$

$$AP = 28 \quad BP = 12; \quad AQ = 70 \quad BQ = 30$$



Ver problema
do Torpedo
Física

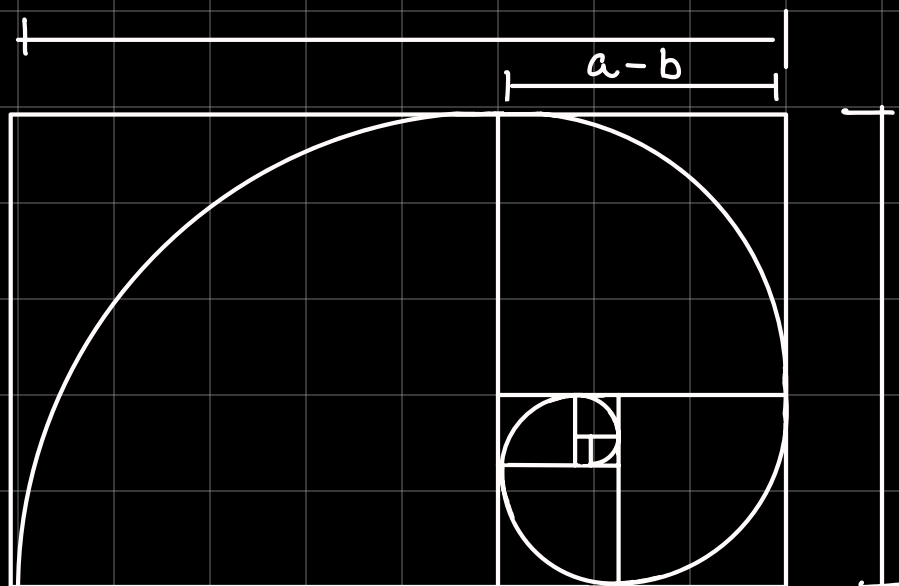


$$\frac{AB}{AP} = \frac{AP}{PB} = \varphi$$

$$\frac{\varphi+1}{\varphi} = \varphi$$

Razão Áurea

$$\frac{a}{b} = \varphi$$



$$\frac{a-b}{b} = \frac{b}{a} \Rightarrow \frac{a}{b} - 1 = \frac{b}{a}$$
$$\varphi - 1 = \frac{1}{\varphi}$$

$$\varphi^2 - \varphi - 1 = 0$$

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618$$

Sofistas $\rightarrow$ Heurística $\rightarrow$ Arte de Argumentar
 $\downarrow$
enrolam $\neq$ Maestica do Séculos
 $\hookrightarrow$ Busca da Verdade

Zenon $\rightarrow$ Paradoxo da Tartaruga de Zenon

$\hookrightarrow$ fez um teste que ele convenceu que aquilo
nunca vai alcançar a Tartaruga

Sequência de Fibonacci ($f_n = f_{n-1} + f_{n-2}$)

$$f_1 = 1, f_2 = 1, f_3 = 2, f_4 = 3, f_5 = 5, f_6 = 8, f_7 = 13$$

$$f_1 = 1, f_2 = 1$$

$$\varphi^2 = \varphi + 1 \Rightarrow \varphi^3 = \varphi^2 + \varphi \Rightarrow \varphi^3 = \underline{2\varphi} + \underline{1}$$

$$\varphi^4 = \varphi^3 + \varphi^2 \Rightarrow \varphi^4 = \underline{3\varphi} + \underline{2}$$

$$\varphi^5 = \varphi^4 + \varphi^3 \Rightarrow \varphi^5 = \underline{5\varphi} + \underline{3}$$

$$\varphi^n = f_n \varphi + f_{n-1}$$

$$\text{Ex IME } \left(\frac{1+\sqrt{5}}{2} \right)^{10} ?$$

$$\varphi^{10} = f_{10} \varphi + f_9$$

$$\varphi^{10} = 55\varphi + 34$$

$$\varphi^{10} = \frac{55\sqrt{5} + 123}{2}$$

$$f_6 = 8$$

$$f_7 = 13$$

$$f_8 = 21$$

$$f_9 = 34$$

$$f_{10} = 55$$

Termo Geral fibonacci

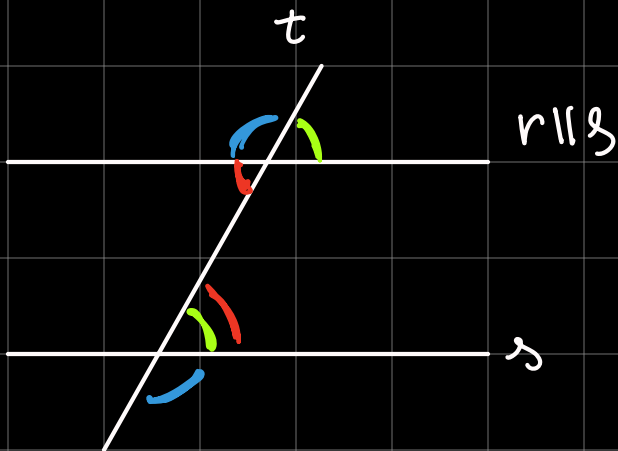
$$\begin{cases} \left(\frac{1+\sqrt{5}}{2} \right)^n = f_n \left(\frac{1+\sqrt{5}}{2} \right) + f_{n-1} \\ \left(\frac{1-\sqrt{5}}{2} \right)^n = f_n \left(\frac{1-\sqrt{5}}{2} \right) + f_{n-1} \end{cases}$$

$\Rightarrow$

$$\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n = \sqrt{5} f_n$$

$$f_n = \frac{\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n}{\sqrt{5}}$$

Paralelismo e Ângulos no Triângulo



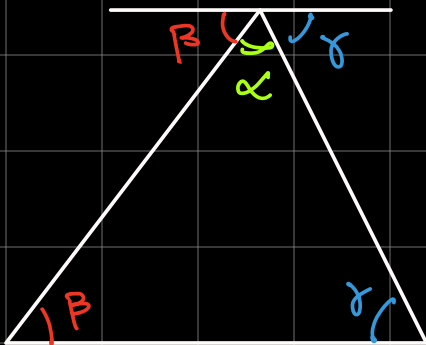
- Correspondentes
- Alternos Internos
- Alternos Externos

teoremas:

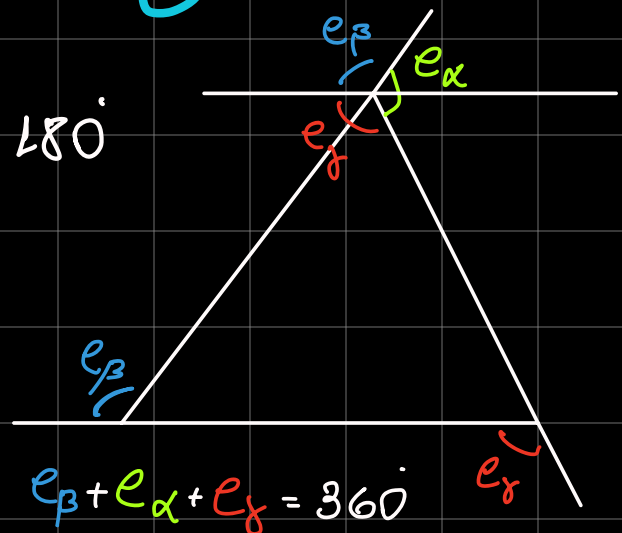
$r \parallel s \iff$ ângulos correspondentes são iguais

$r \parallel s \iff$ ângulos alternos internos são iguais

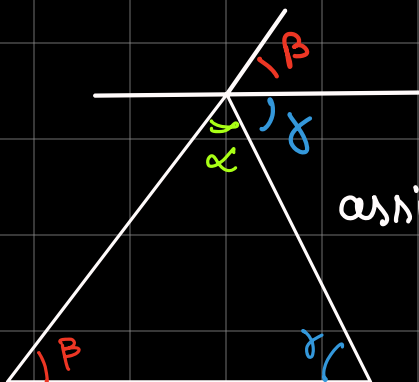
Ângulos no Triângulo



$$\alpha + \beta + \gamma = 180^\circ$$



$$e_\beta + e_\alpha + e_\gamma = 360^\circ$$



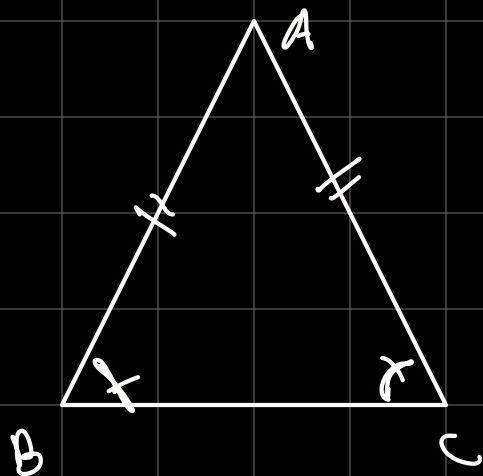
assim

$$e_\alpha = \beta + \gamma$$

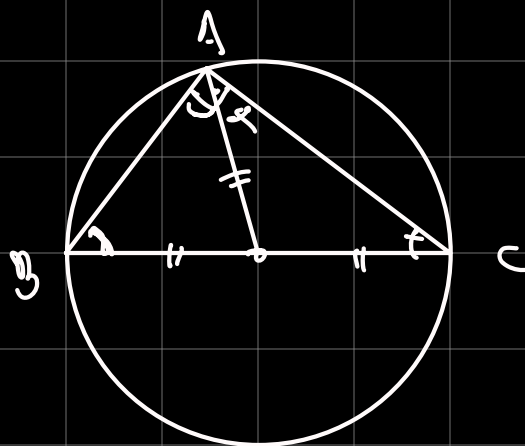
Ver a Janela do
Euclides, Leonard
Euler

Teorema do Isóscelo

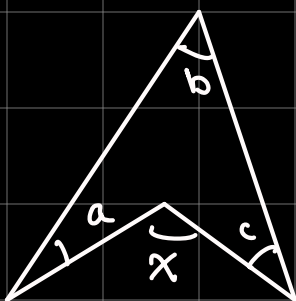
→ elemento do fey n man



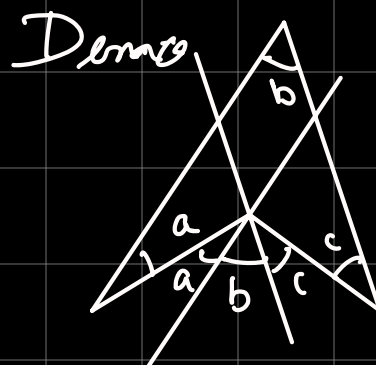
$$AB = AC \Leftrightarrow \hat{B} = \hat{C}$$



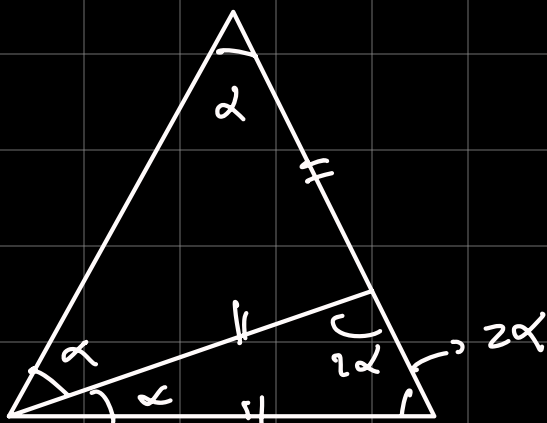
Quadrilátero Côncavo



$$x = a + b + c$$

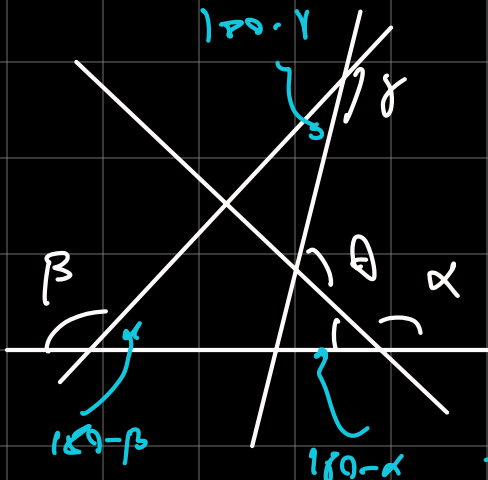


Exercício 12



$$\begin{aligned} 5\alpha &= 180 \\ \alpha &= 36^\circ \end{aligned}$$

Good filters converge



$$3 \cdot (150) - \alpha - \beta - \gamma = 0$$

$$540 - (\alpha + \beta + \gamma) = \theta$$

$$b_0 = \alpha_T \beta + \gamma + \theta$$

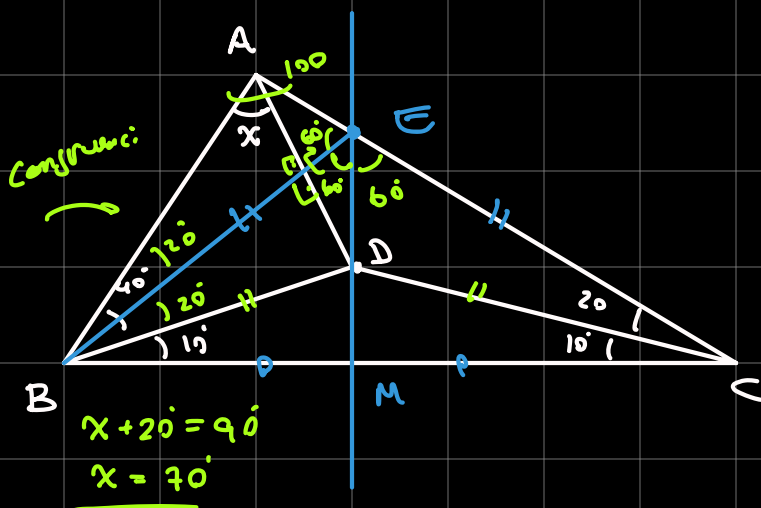
$$\begin{array}{r} 2150 \\ 3 \\ \hline 540 \end{array}$$
[illegible]

o mediano
velho método
de hipotenusa
do triângulo
retângulo?
Sempre?

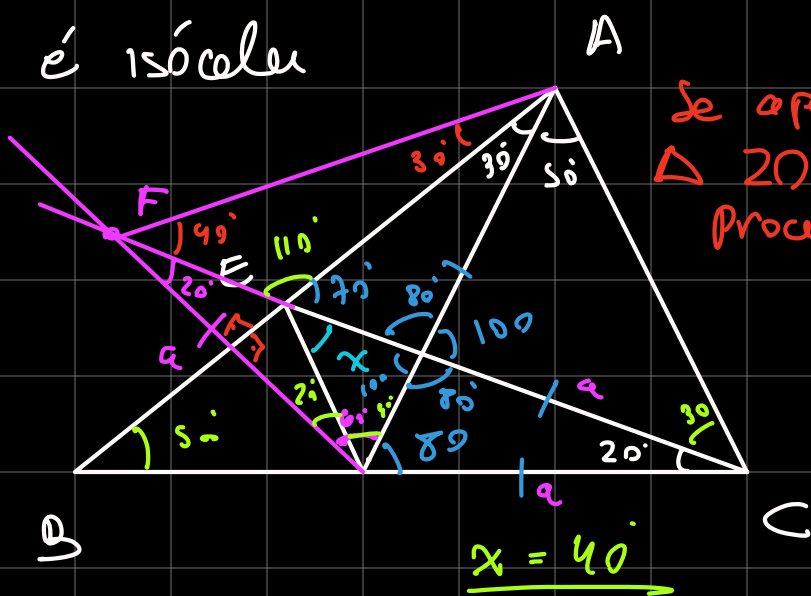
90-26° 64' 64 → 90

$$Z = Si + 26 \quad Z = 78$$

451

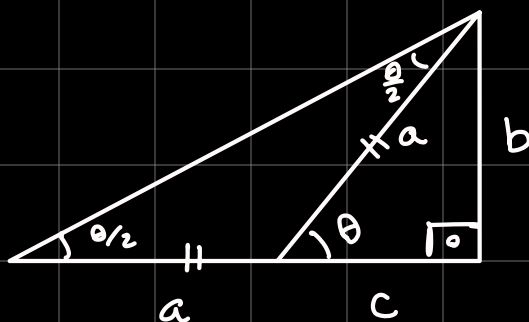


42 é isóceles



Se aparecer
 $\Delta 20^\circ 80^\circ 80^\circ$
 procure equilátero
 =

Divisor do Ângulo



$$\operatorname{tg} \frac{\theta}{2} = \frac{b}{a+c} \quad \frac{\div a}{\div a} \quad \frac{\frac{b}{a}}{1+\frac{c}{a}} = \frac{\sin x}{1+\cos x}$$

- Se aparecer $\Delta 20^\circ, 80^\circ, 80^\circ$ procurar equilátero

Congruência de Triângulos

LAL



LLL



ALA



HC



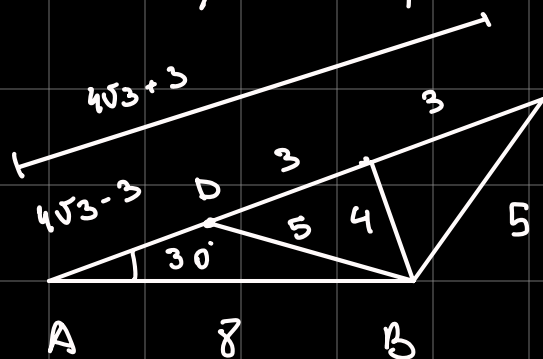
(LLA₀ → Não é Congruência)
Só Neste Caso

LLA₀



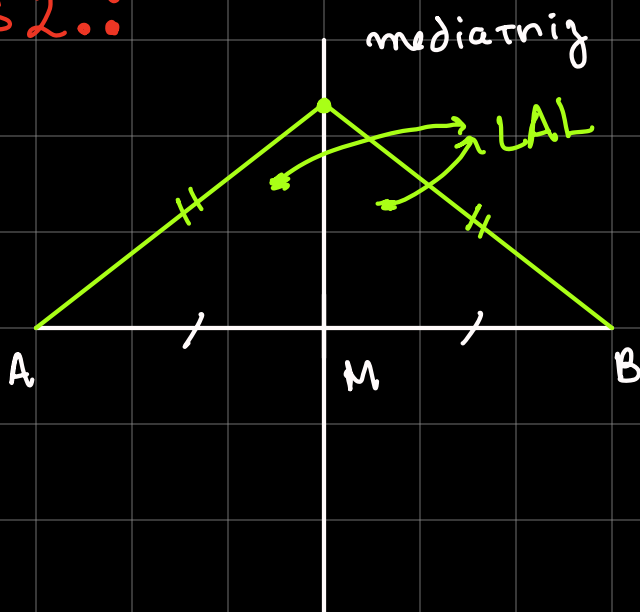
obs 1.: LLA₀ não é caso de congruência

$$AB = 8, BC = 5, \hat{BAC} = 30^\circ$$



TANTO
 $\triangle ABD$ e $\triangle ABC$
obedecem LLA₀ e não congruentes

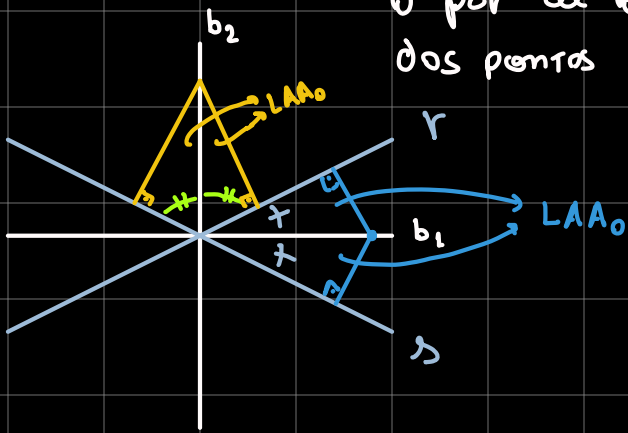
obs 2.:



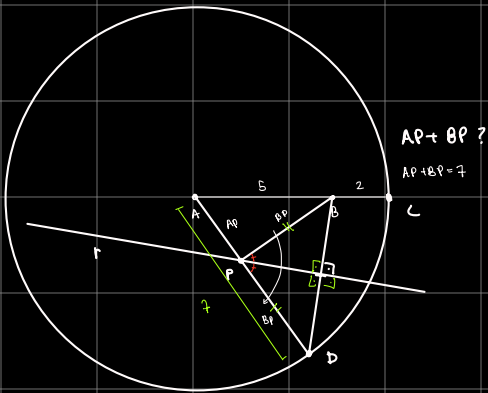
Qualquer ponto na mediatriz de $\overline{AB}$ é L.G.
os pontos que equidistam A e B

obs 3.:

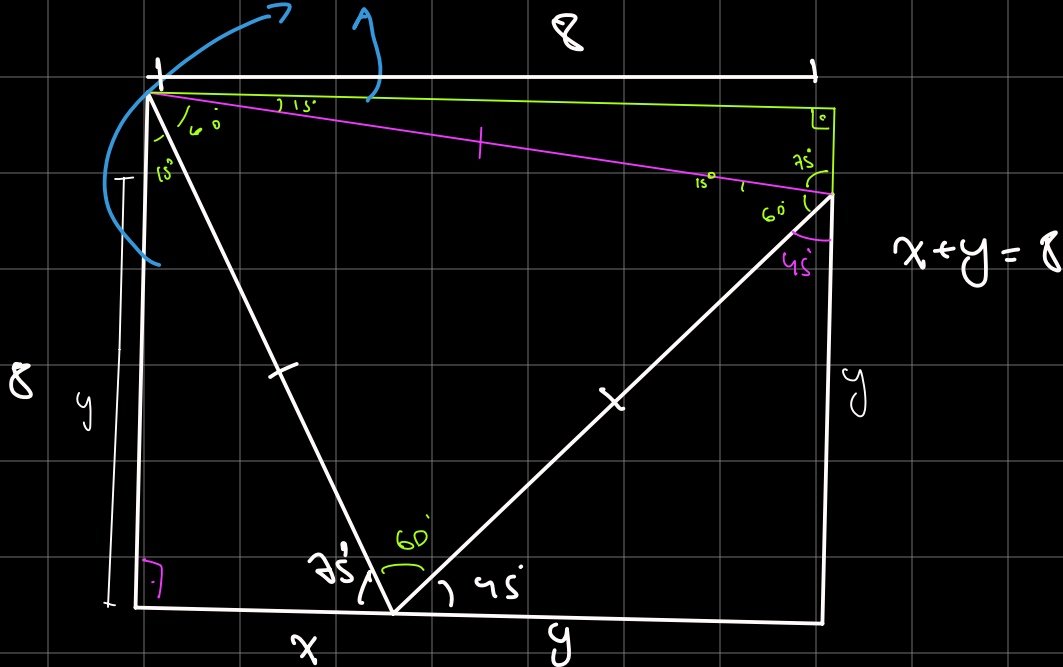
O par de bissetrizes de r, s é o LG dos pontos do plano que equidistam de r e s



Exercício

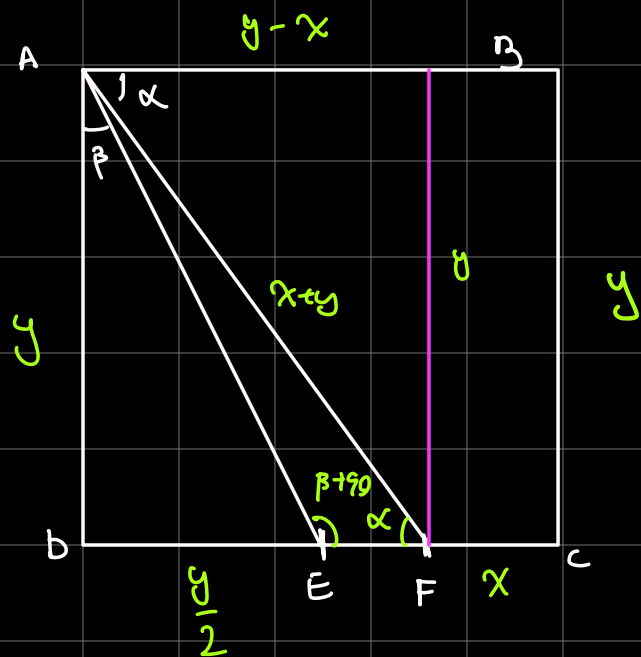


Exercício 8 ALA



Exercício

$$S = \int \frac{p^2}{2m} dt - \int v dt$$



$$\cos \alpha = \frac{y-x}{y+x}$$

$$\cos \beta = \frac{y}{\frac{y\sqrt{5}}{2}}$$

$$\cos \beta = \frac{2\sqrt{5}}{5}$$

$$\cos \alpha = \frac{3}{5}$$

$$2\cos^2 \beta - 1$$

$$(y-x)^2 + y^2 = (x+y)^2$$

$$y - 2x = 2x$$

$$y^2 + x^2 - 2xy + y^2 = x^2 + y^2 + 2xy$$

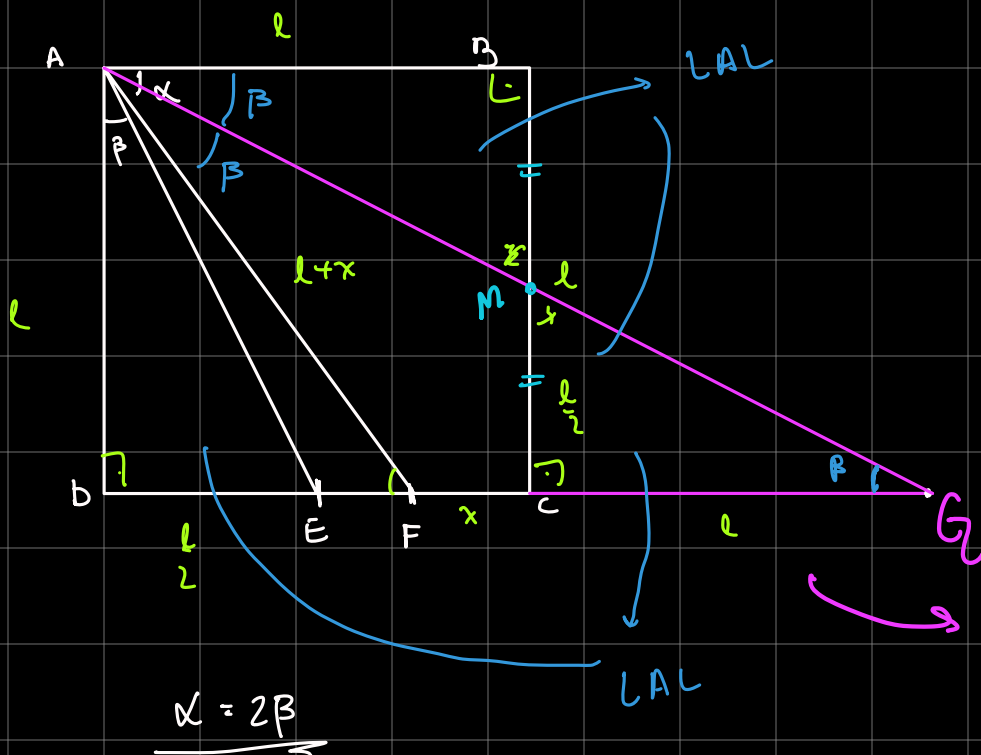
$$y = 4x$$

$$2 \cdot \frac{4 \cdot 5}{2 \cdot 5} - 1$$

$$\frac{8}{5} - \frac{5}{5} = \frac{3}{5}$$

Construção

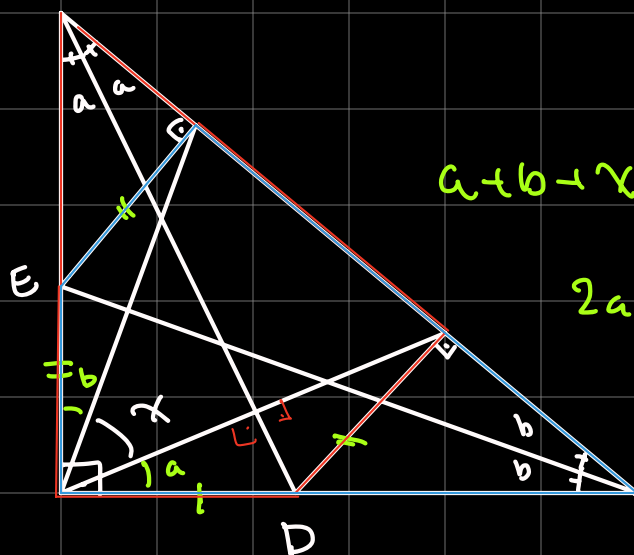
1. 2. 3. 4.



Forçar Isóceles

$$\alpha = 2\beta$$

Exercício

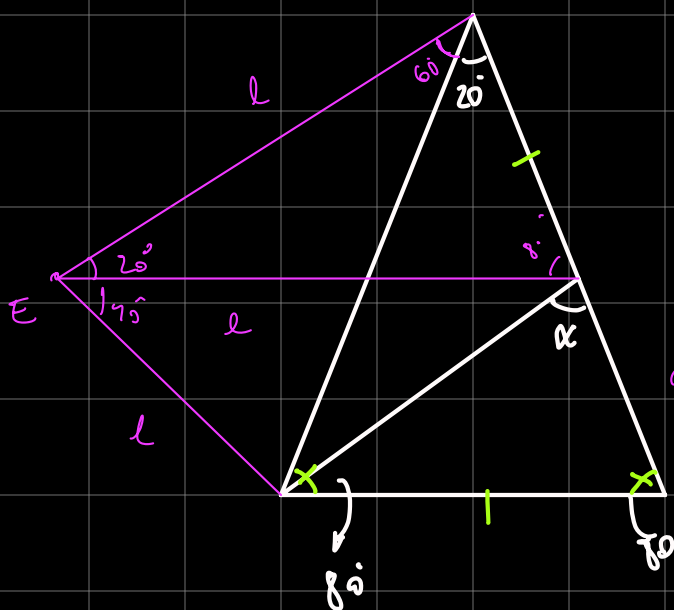


$$a + b + x = 90^\circ$$

$$2a + 2b = 90$$

$$x = 45^\circ$$

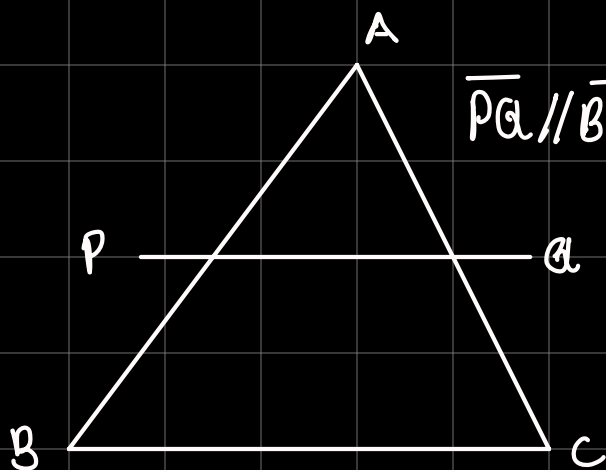
Exercício 37



$$40^\circ + 70^\circ + 80^\circ = 190^\circ$$

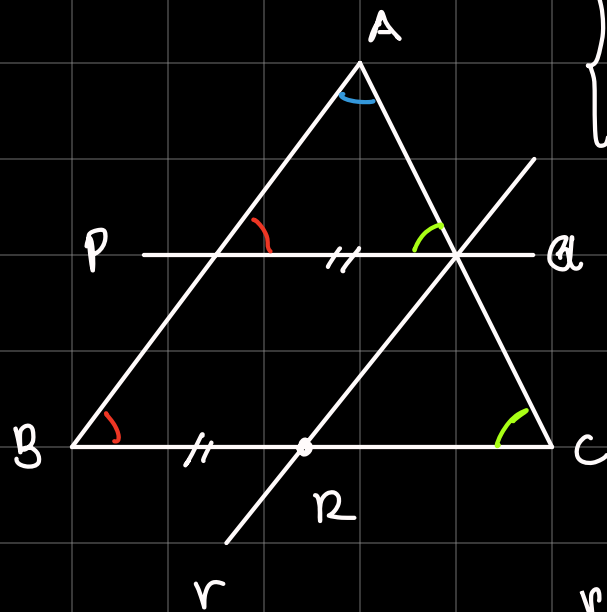
$$\alpha = 30^\circ$$

Teorema Fundamental da Semelhança



$$\overline{PQ} \parallel \overline{BC} \Rightarrow \triangle APQ \sim \triangle ABC$$

Demonstração



$$(i) \begin{cases} \hat{P}\hat{A}\hat{Q} = \hat{B}\hat{A}\hat{C} \text{ (common)} \\ \hat{A}\hat{P}\hat{Q} = \hat{A}\hat{B}\hat{C} \text{ (corresponding)} \\ \hat{A}\hat{Q}\hat{P} = \hat{A}\hat{C}\hat{B} \text{ (corresponding)} \end{cases}$$

(iii) Tales as Transversais

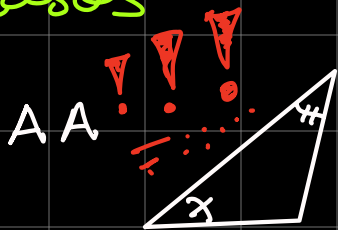
$$\frac{AP}{AB} = \frac{AQ}{AC} *$$

$$\text{em } PQ \text{ e } BR \quad \overline{PA} = \overline{BR}$$

$r \parallel AB$, tal as transversais

$$\frac{BR}{BC} = \frac{AQ}{AC} \Rightarrow \frac{PQ}{BC} = \frac{AQ}{AC} \quad \swarrow$$

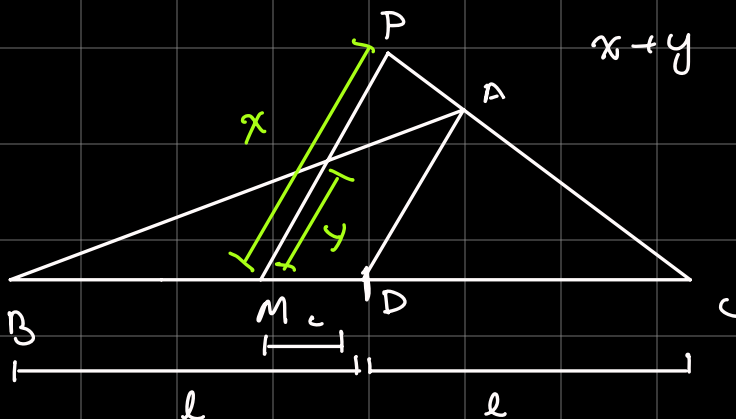
Casos



LLL só precisam ser proporcionais os lados



Exercício



$x+y$

$$\triangle ADC \sim \triangle MPC$$

$$\frac{AD}{x} = \frac{l}{l+c}$$

$$\frac{y}{AD} = \frac{l-c}{l}$$

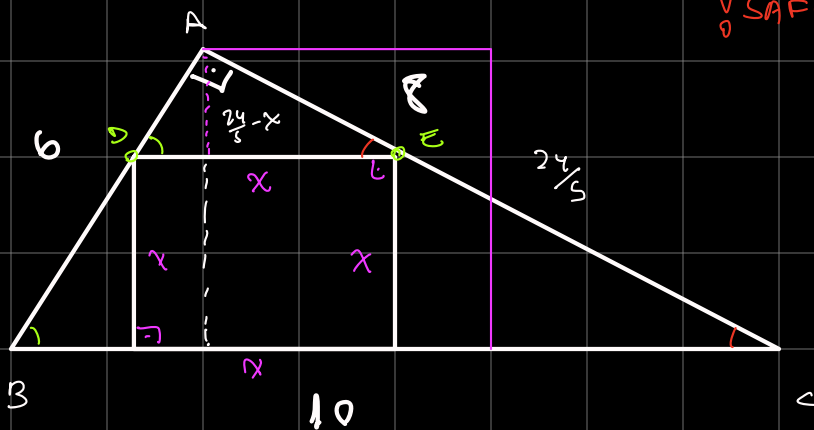
$$ly = AD(l-c)$$

$$+ lx = AD(l+c)$$

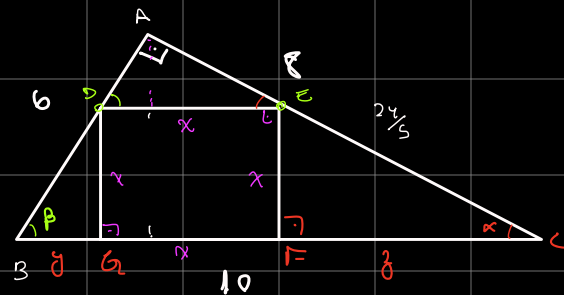
$$l(x+y) = 2ADl$$

$$x+y = 2AD$$

Exercício 14



$\nabla_{SAFADIC} \text{ ITAUC } \Delta \approx 3,4,5$



$\Delta ABC \sim \Delta GDB$ e $\Delta EFC \sim \Delta ABC$

$$TS \alpha = \frac{5}{8} \quad TS \beta = \frac{5}{6}$$

$$y + x + z = 10 \Rightarrow x TS \beta + x + x TS \alpha = 10$$

$$x (TS \beta + TS \alpha + 1) = 10$$

$$\frac{24}{5}$$

$$\left(\frac{3}{4} + \frac{4}{5} + \frac{12}{12} \right)$$

$$\left(\frac{9+16}{12} + \frac{12}{12} \right)$$

$$x \frac{37}{12} = 10 \quad x = \frac{120}{37}$$

$$\frac{6 \cdot 8}{2} = \frac{10 \cdot (h+x)}{2} = \frac{24}{5} = h+x$$

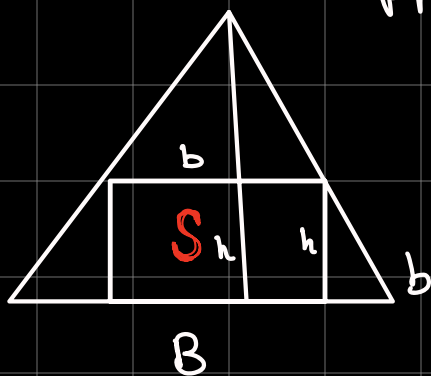
$\Delta ABC \sim \Delta ADE$ (AA)

$$\frac{x}{10} = \frac{24-x}{6} \Rightarrow \frac{24x}{5} = 48 - 10x$$

$$\frac{74x}{5} = 48$$

$$\frac{37x}{5} = 24 \quad x = \frac{24 \cdot 5}{37} = \frac{120}{37}$$

Exercício 17



Prova $\frac{b}{B} + \frac{h}{H} = 1$

$$\frac{b}{B} = \frac{H-h}{H} \Rightarrow \frac{b}{B} - \frac{H}{H} + \frac{h}{H} = 0$$

$$\frac{b}{B} + \frac{h}{H} = 1$$

b) Prova que Área Máxima $b = \frac{B}{2}, h = \frac{H}{2}$

$$\frac{b}{B} + \frac{h}{H} = 1 \Rightarrow h = \left(1 - \frac{b}{B}\right) H \rightarrow \text{linear}$$

$$S = b \cdot h \Rightarrow \left(1 - \frac{b}{B}\right) H b$$

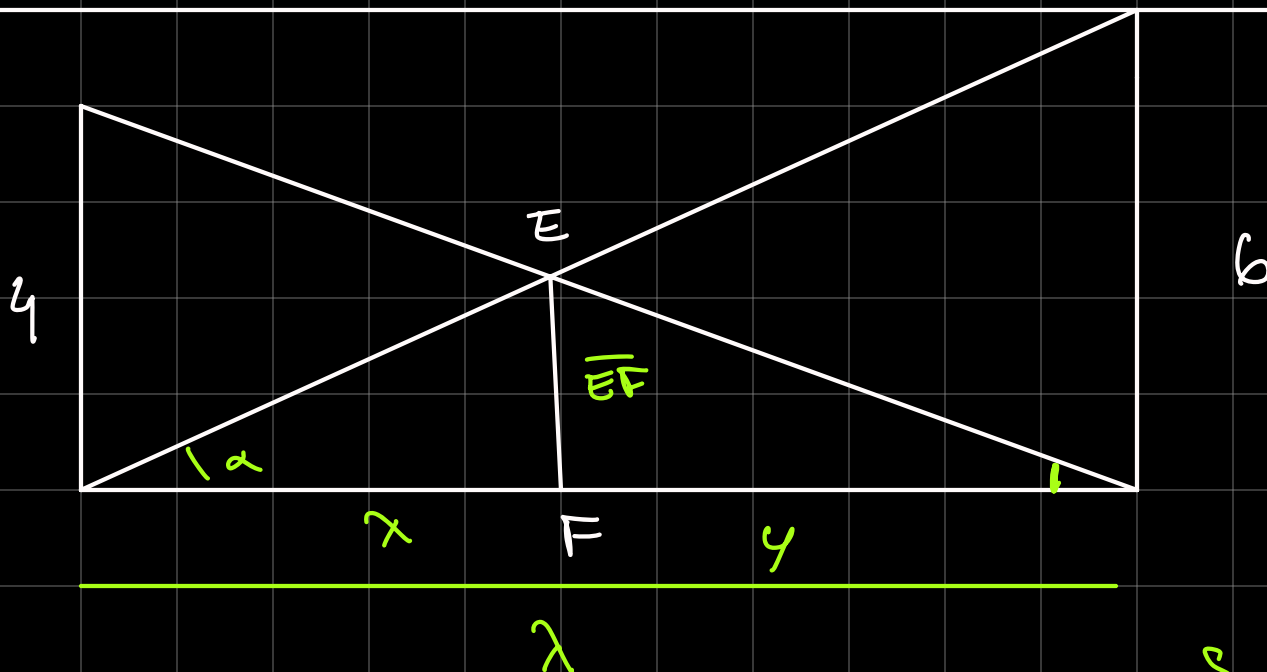
$$S = \left(b - \frac{b^2}{B}\right) H \quad \text{max}$$

$$H \frac{d(b - \frac{b^2}{B})}{db} = 0$$

$$H(1 - \frac{2b}{B}) = 0 \Rightarrow B = 2b \quad b = \frac{B}{2}$$

$$\cancel{\frac{B}{2}} + \frac{h}{H} = 1 \Rightarrow \frac{h}{H} = \frac{1}{2} \Rightarrow h = \frac{H}{2}$$

Area Máxima é $\frac{BH}{4}$ que é $\frac{SA}{2}$



$$\frac{4}{x+y} = 1$$

$$\frac{b}{x+y} = 1$$

$$\frac{2}{3}y = x$$

$$\frac{EF}{x} = \frac{b}{x+y}$$

$$\frac{EF}{y} = \frac{4}{x+y}$$

$$x EF + y EF = bx$$

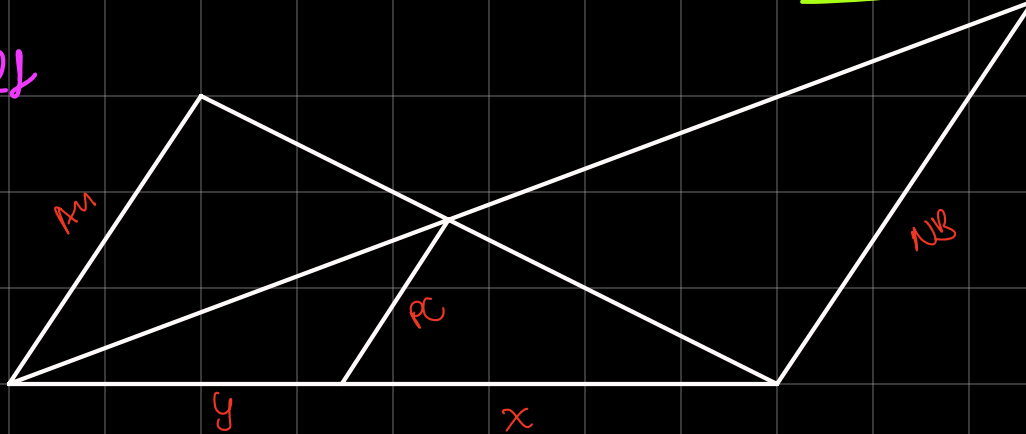
$$x EF + y EF = 4y$$

$$x = \frac{2}{3}y$$

$$\frac{EF}{4} = \frac{y}{\frac{2}{3}y}$$

$$\frac{12}{5} = EF$$

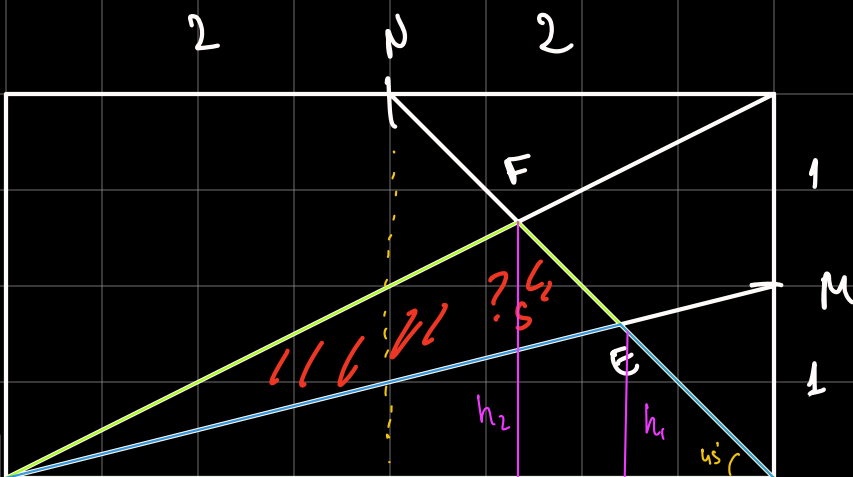
Exercício 21



$$\left[\begin{array}{l} \frac{x}{PC} = \frac{x+y}{AM} \\ + \\ \frac{y}{PC} = \frac{x+y}{NB} \end{array} \right]$$

$$\frac{x+y}{PC} = \frac{x+y}{AM} + \frac{x+y}{NB} \Rightarrow \frac{1}{PC} = \frac{1}{AM} + \frac{1}{NB}$$

Exercício 33



S?

com replicando 1 base
do mesmo

$$S_2 - S_1 = S$$

$$\frac{1}{h_2} = \frac{1}{2} + \frac{1}{4} \Rightarrow h_2 = \frac{4}{3}$$

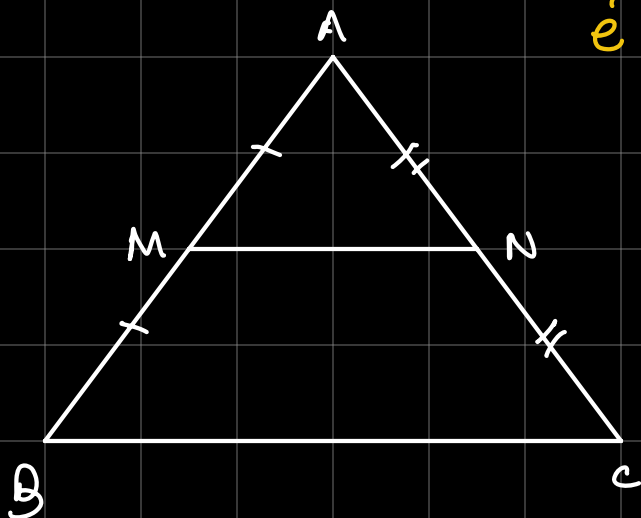
$$\frac{1}{h_1} = \frac{1}{4} + \frac{1}{1} \Rightarrow h_1 = \frac{4}{5}$$

$$S = 4 \left(\frac{4}{3} - \frac{4}{5} \right) \frac{1}{2}$$

$$S = \left(\frac{20 - 12}{15} \right) 4 = \frac{16}{15}$$

Base Média

1) Do Triângulo



é notável que

- $MN \parallel BC$
- $MN = \frac{BC}{2}$

$$\text{Logo } \frac{AM}{AB} = \frac{1}{2} = \frac{AN}{AC}$$

$$\hat{M}AN = \hat{B}AC$$

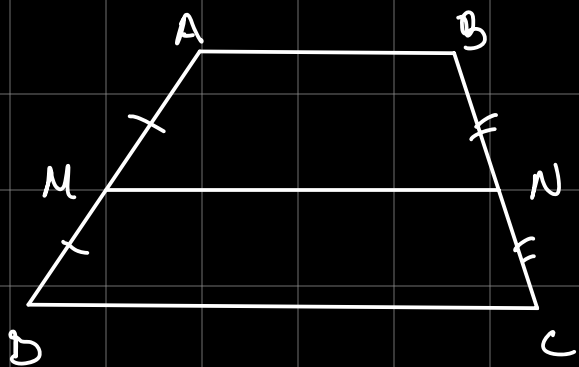
$\Downarrow$ LAL

$$\triangle AMN \sim \triangle ABC$$

$$MN \parallel BC \leftarrow$$

$$\frac{MN}{BC} = \frac{AM}{AC} = \frac{1}{2} \Rightarrow MN = \frac{BC}{2}$$

2) Trapézio

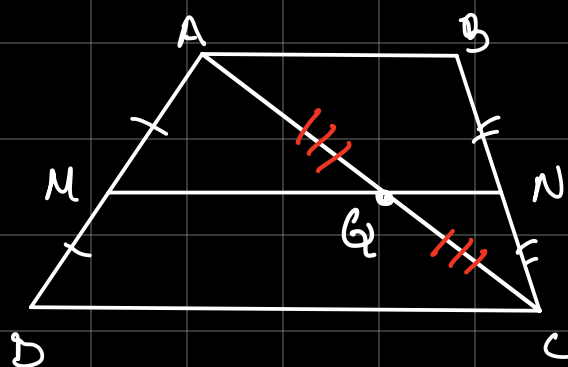


Notável que:

$$\bullet AB \parallel MN \parallel CD$$

$$\bullet MN = \frac{AB + CD}{2}$$

Demo: Como $\overline{AB} \parallel \overline{CD}$ e $\frac{AM}{MD} = \frac{BN}{NC} = 1$ pela
recíproca de Tales $\overline{MN} \parallel \overline{AB} \parallel \overline{CD}$



$$\frac{AG}{GC} = \frac{AM}{MD} = 1 \Rightarrow AG = GC$$

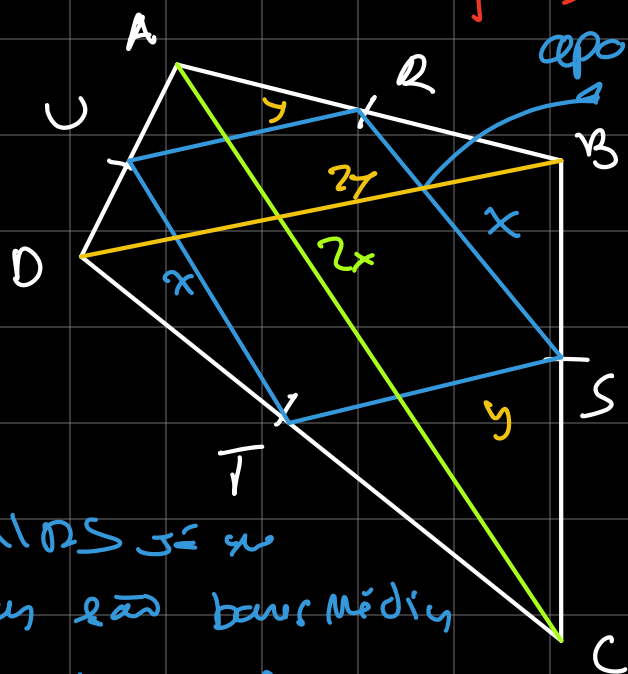
analogamente GN é base média de $\triangle ADC$
e GN é base média de $\triangle ABC$

$$MG = \frac{DC}{2}; \quad GN = \frac{AB}{2}$$

$$MN = \frac{AB + DC}{2}$$

Exercício 01: Qual o perímetro do $\square RSTU$
em função das diagonais

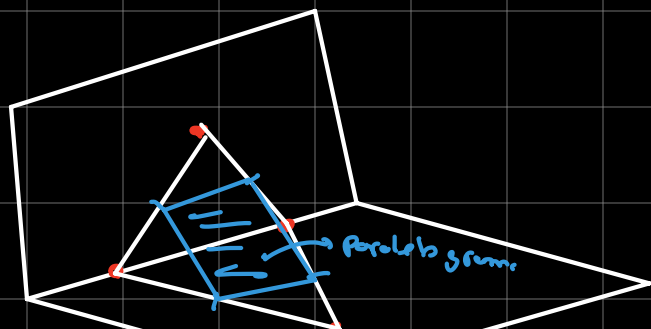
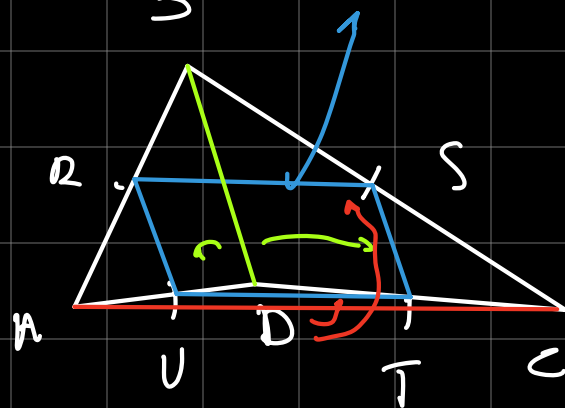
aproveite um paralelogramo



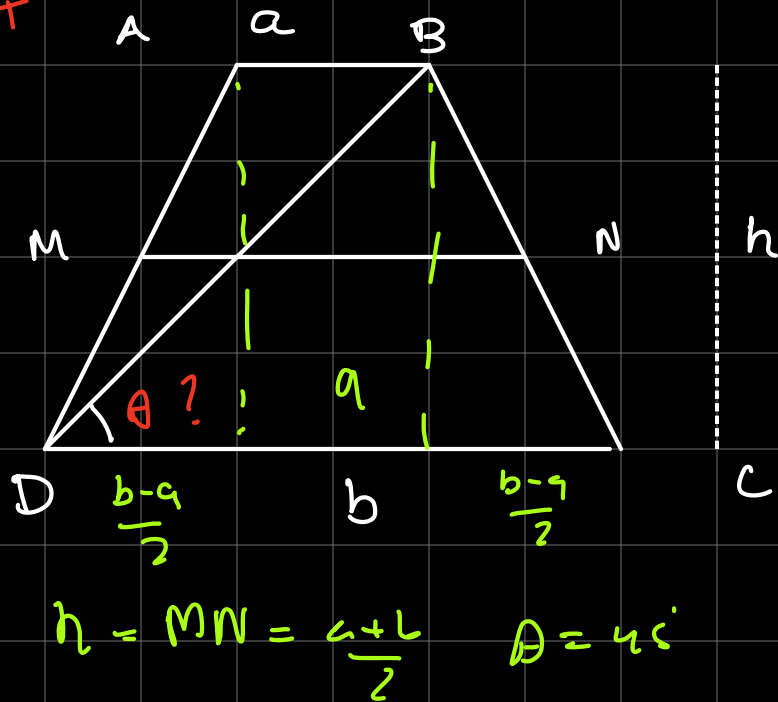
$DT \parallel RS$ e
ambas são bases médias
referentes a AC

$$RSTU \text{ IP} = x + x + y + y$$

$$2P = 2x + 2y = AC + BD$$

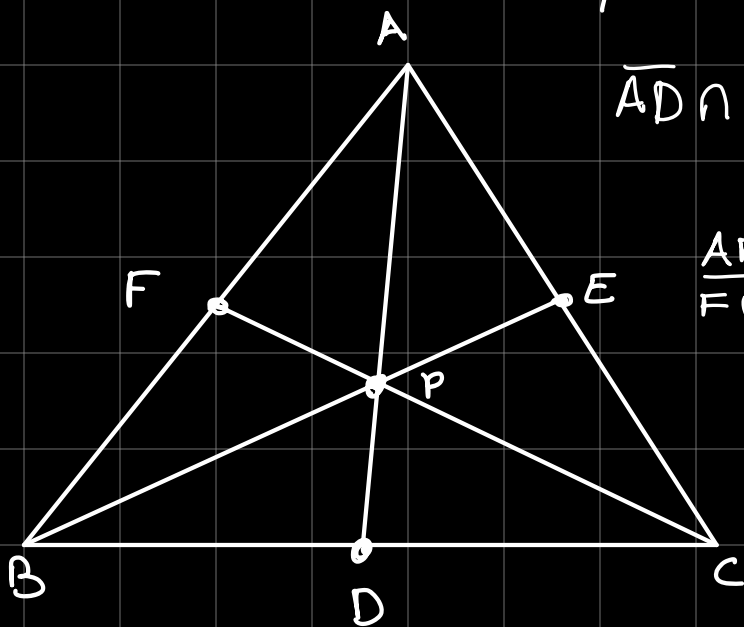


Ex 3 Fuvest
Trapezoido e'
Isosceles



Teorema de Ceva Augustin Ceva

∀ Triangulo, Mostrar que concorrem no mesmo ponto

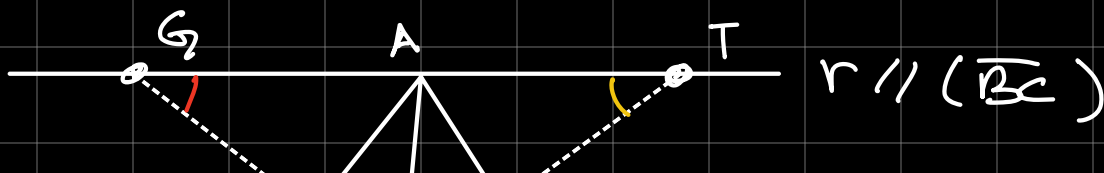


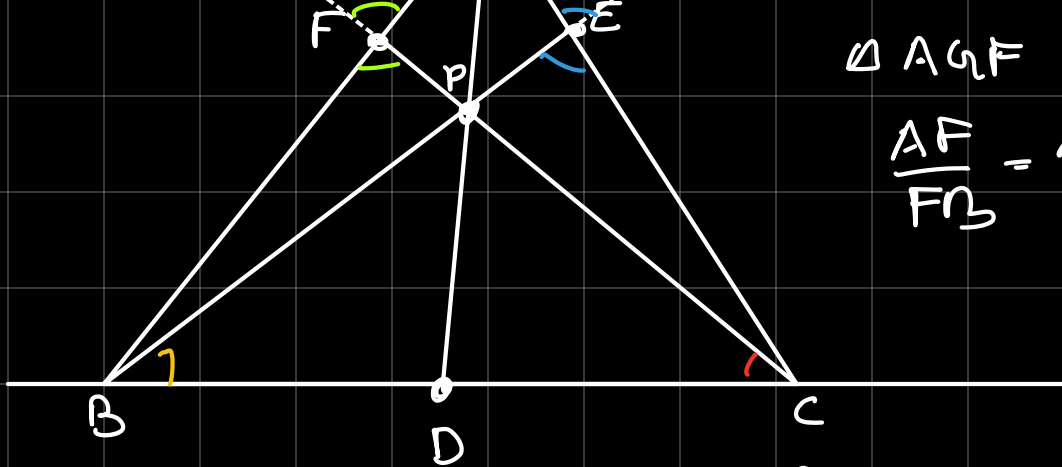
$$\overline{AD} \cap \overline{BE} \cap \overline{CF} = \{P\}$$

$$\iff \frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1$$

Concorrem
em um mesmo
ponto

Demo: usando semelhança





$$\triangle AGF \sim \triangle BCF$$

$$\frac{AF}{FB} = \frac{AG}{GC}$$

$$\triangle BCE \sim \triangle ATE : \frac{CE}{EA} = \frac{BC}{AT}$$

$$\triangle BDP \sim \triangle ATP : \frac{BD}{AT} = \frac{DP}{PA}$$

$$\triangle CDP \sim \triangle ASP : \frac{CD}{AS} = \frac{DP}{PA}$$

$$\frac{BD}{CD} = \frac{AT}{AS}$$

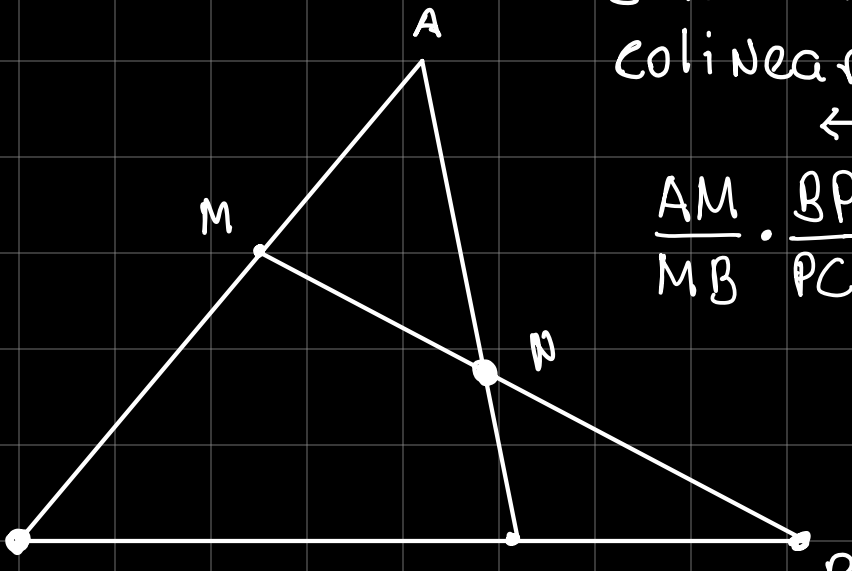
$$\frac{AS}{BC} \cdot \frac{AT}{AB} \cdot \frac{BC}{AT} = \frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1$$

Teorema de Menelaus

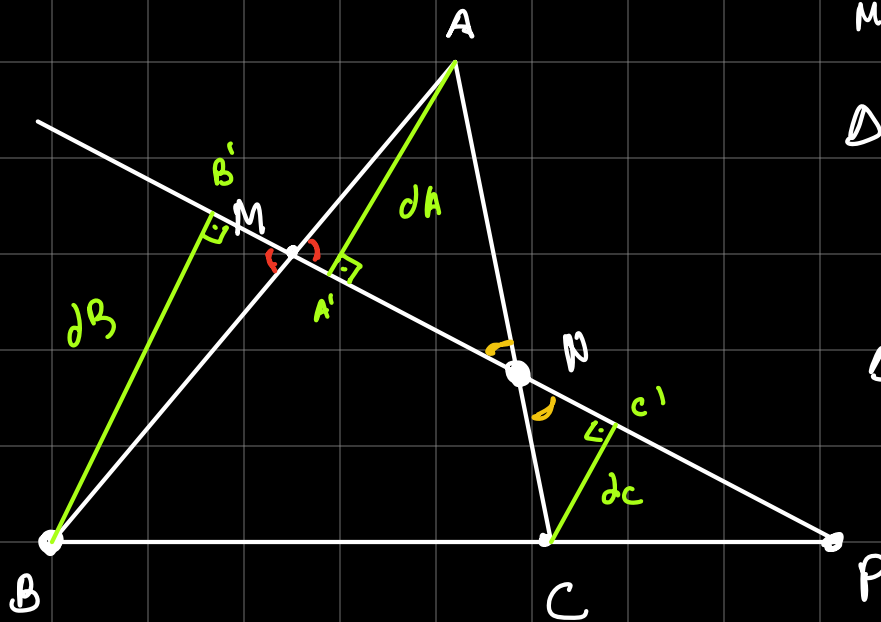
Seja M, N, P
colineares



$$\frac{AM}{MB} \cdot \frac{BP}{PC} \cdot \frac{CN}{NA} = 1$$



Demo:



$$\triangle A'AM \sim \triangle BB'M$$

$$\frac{AM}{MB} = \frac{dA}{dB}$$

$$\triangle BB'P \sim \triangle CC'P$$

$$\frac{BP}{PC} = \frac{dB}{dC}$$

$$\triangle CC'N \sim \triangle AA'N$$

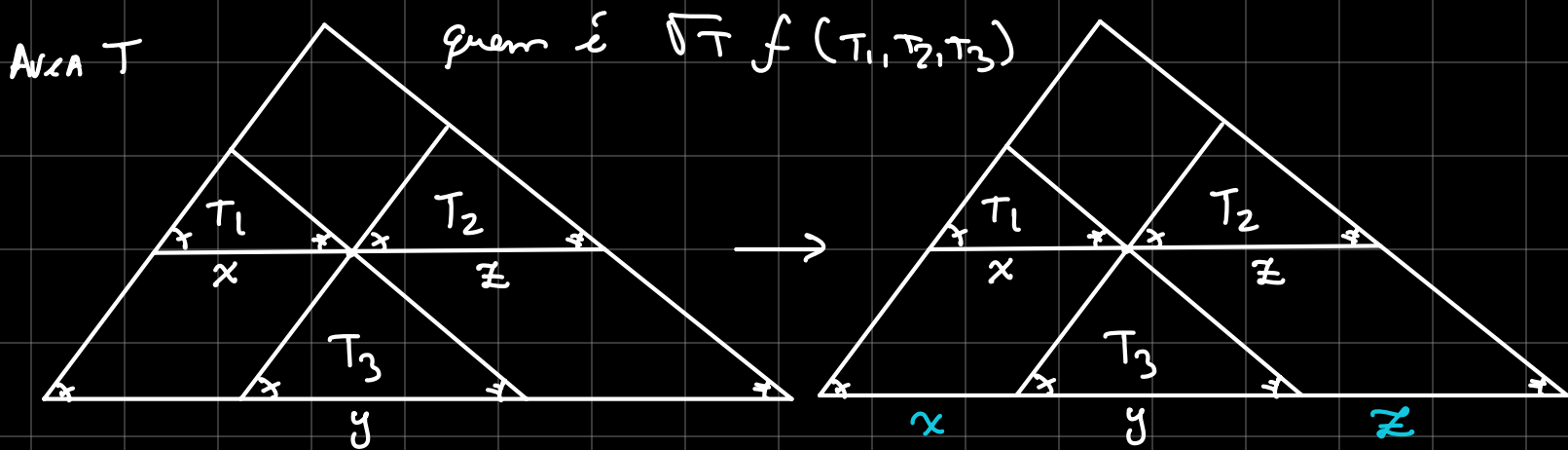
$$\frac{CN}{AN} = \frac{dC}{dA}$$

$$\frac{AM}{MB} \cdot \frac{BP}{PC} \cdot \frac{CN}{NA} = \frac{dA}{dB} \cdot \frac{dB}{dC} \cdot \frac{dC}{dA} = 1$$

Razão entre Áreas

1) Figuras Semelhantes

- Se as duas figuras são semelhantes e a razão entre os comprimentos é k , então a razão entre áreas é k^2



$$\frac{T_1}{T} = \left(\frac{x}{x+y+z} \right)^2 ; \quad \frac{T_2}{T} = \left(\frac{y}{x+y+z} \right)^2 ; \quad \frac{T_3}{T} = \left(\frac{z}{x+y+z} \right)^2$$

$$\sqrt{T_1} = \sqrt{T} \left(\frac{x}{x+y+z} \right)$$

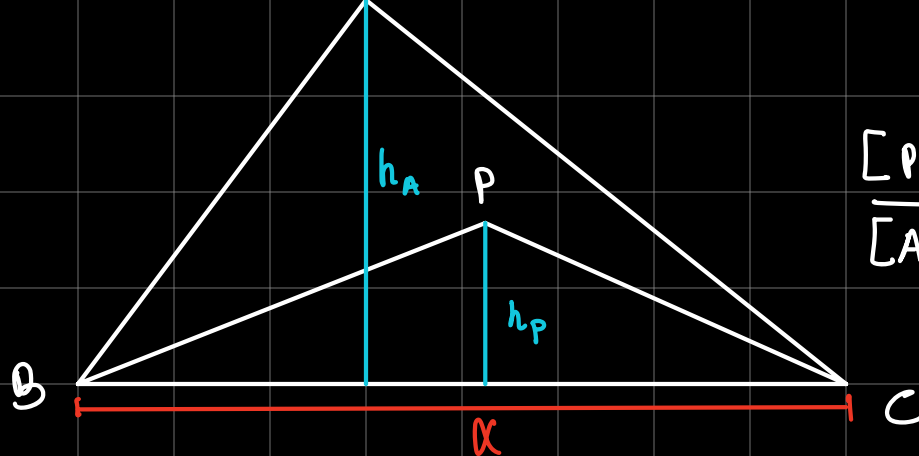
$$\sqrt{T_2} = \sqrt{T} \left(\frac{y}{x+y+z} \right)$$

$$\sqrt{T_3} = \sqrt{T} \left(\frac{z}{x+y+z} \right)$$

$$\sqrt{T} \left(\frac{x+y+z}{x+y+z} \right) = \sqrt{T_1} + \sqrt{T_2} + \sqrt{T_3}$$

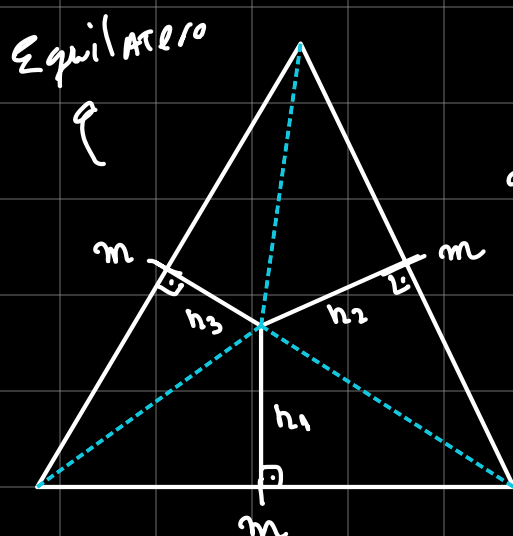
$$\sqrt{T} = \sqrt{T_1} + \sqrt{T_2} + \sqrt{T_3}$$

2) Triângulos que Compartilham 1 Lado



$$\frac{[PBC]}{[ABC]} = \frac{a \cdot h_P/2}{a \cdot h_A/2} = \frac{h_P}{h_A}$$

Exemplo Determine $h_1 + h_2 + h_3$

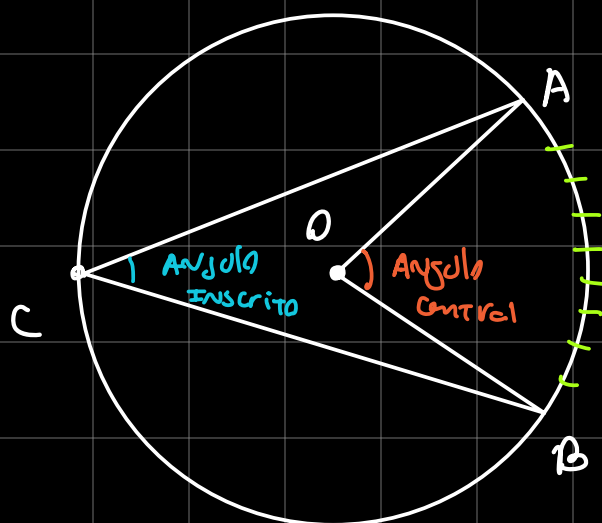


$$A = A_1 + A_2 + A_3$$

$$\frac{m^2 \sqrt{3}}{4} = \frac{m h_1}{2} + \frac{m h_2}{2} + \frac{m h_3}{2}$$

$$\frac{m \sqrt{3}}{2} = h_1 + h_2 + h_3$$

Ângulos na Circunferência



Def: Medida Angular do Arco

$$m(\widehat{AB}) = m(\widehat{AOB})$$

→ Centro

Teorema do Ângulo Inscrito.

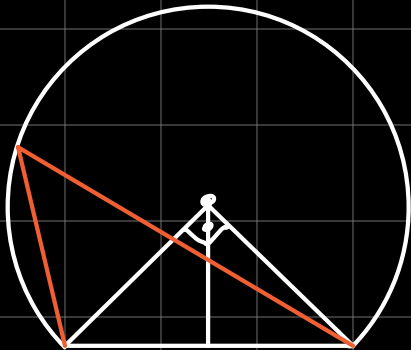
O ângulo inscrito que intercepta a mesma arco do central

ГЛАСНОСТЬ

e motech on Central

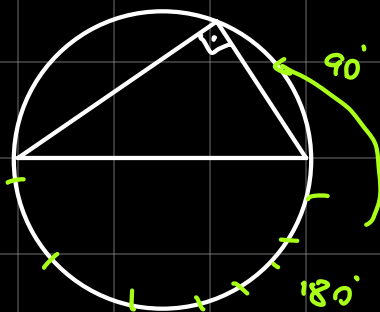
$$\text{Angulo Inscrito} = \frac{1}{2} \text{Angulo Central}$$

Arco Copo

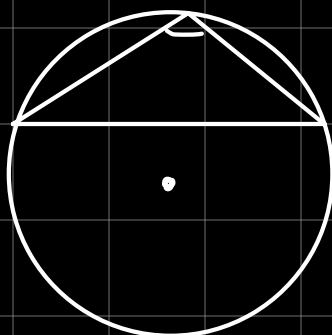


Todos pontos sobre
essa Circunferencia
Tiram fora com
Angulo de 45° do arco

Consequencias

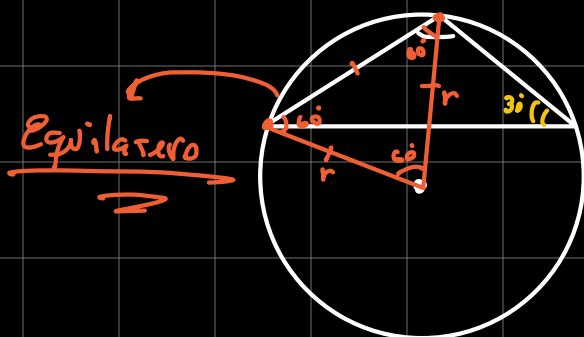


A Circ é o l-G
dos vertices do Triangulo
Retangulo de hipotenusa e
tamanho do diametro

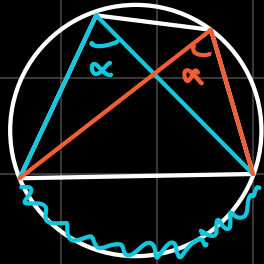
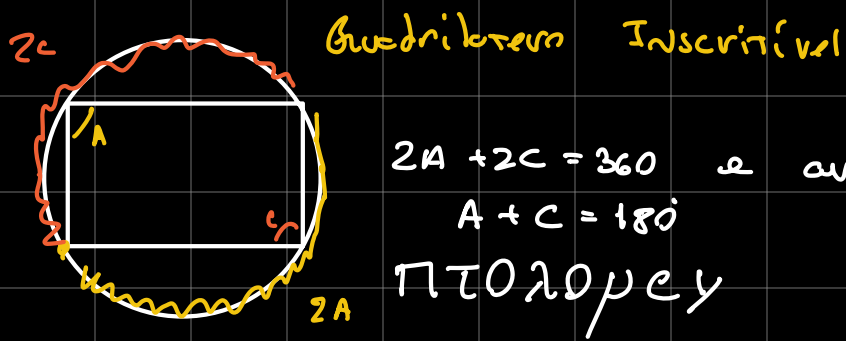


OBTUSANGULOS tem o
Centro da Circunferencia Circunscrita
Fora

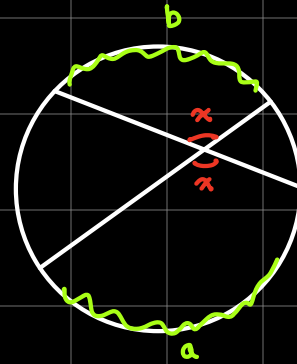
Se tivermos um obtusangulo de angulo de 30°



desenho lado oposto ao
30°

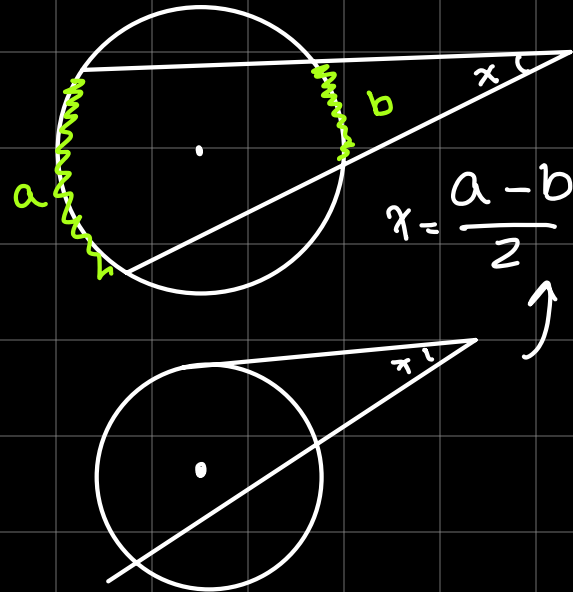


Geometria Interior

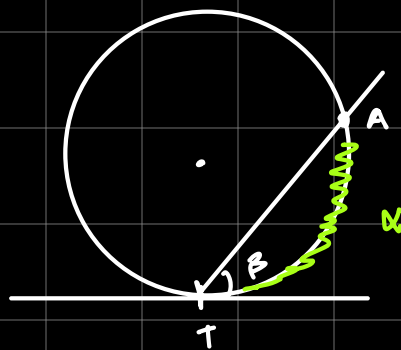


$$x = \frac{a+b}{2}$$

Geometria Exterior



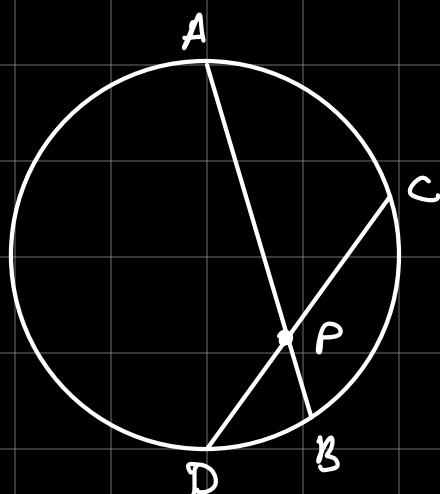
ângulo de segmento



$$x = \frac{1}{2}r$$

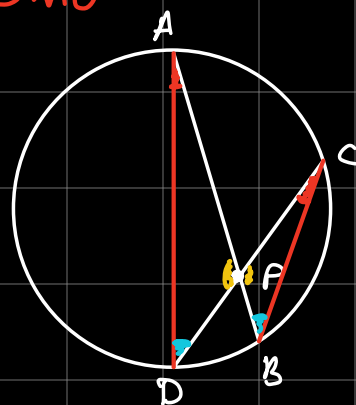
Relações Métricas Da Circunferência.

1.) Teorema das Cordas



$$PA \cdot PB = PC \cdot PD$$

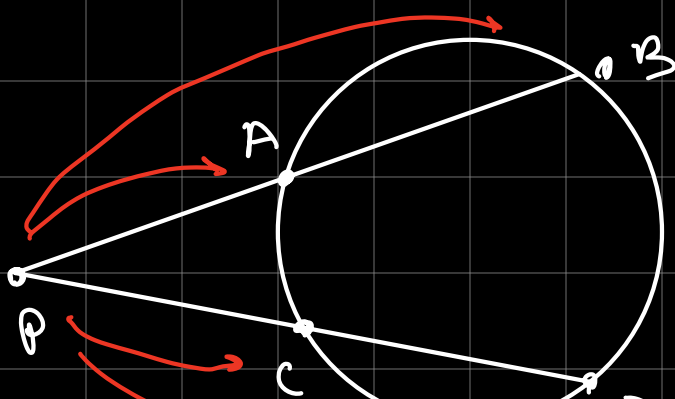
Demo



$\triangle PAD \sim \triangle PCB$ por AAA

$$\frac{PA}{PC} = \frac{PD}{PB} \rightarrow PA \cdot PB = PC \cdot PD$$

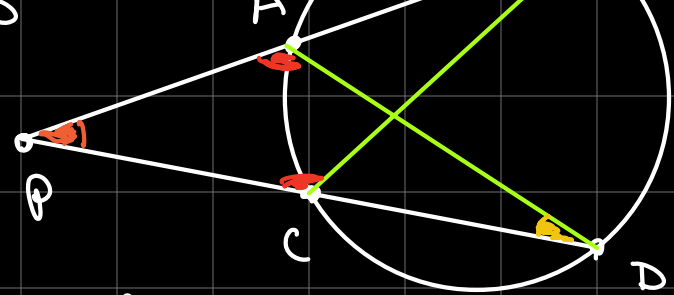
2.) Teorema das Secantes



$$PA \cdot PB = PC \cdot PD$$

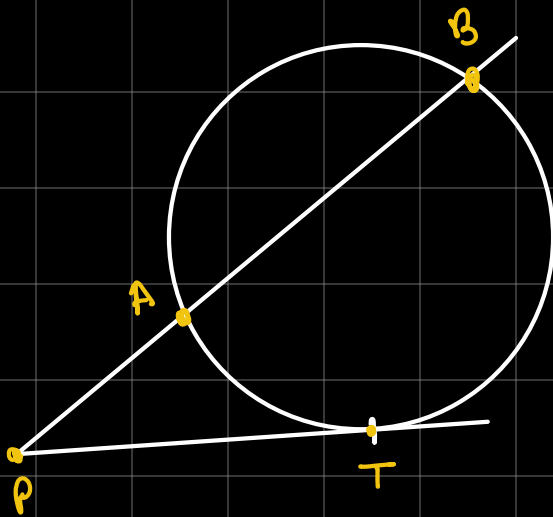
Demo:





$$\frac{PA}{PC} = \frac{PD}{PB} \Rightarrow PA \cdot PB = PD \cdot PC$$

3) Teorema da Tangente - Secante



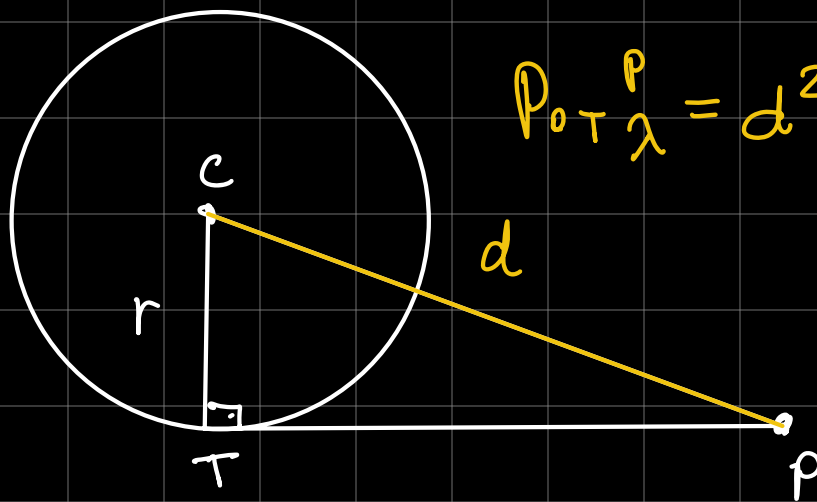
$$PA \cdot PB = PT^2$$

Demo: Caso limite do Teorema da Secante

Perceba que os 3 teoremas são na verdade descrições de mesmo fenômeno

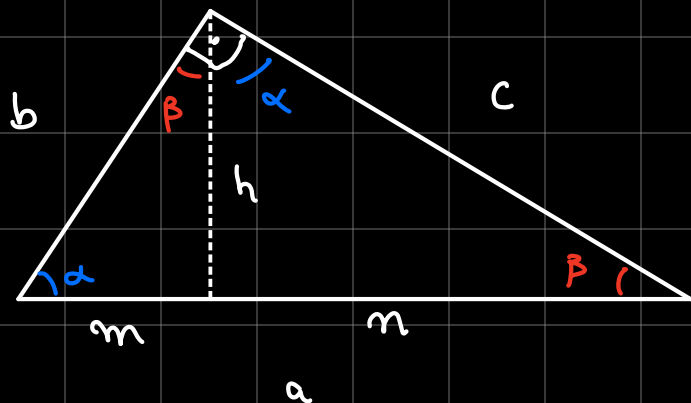
Perceba tbm que independentemente de A e B eles sempre guardarão uma razão fixa entre si que só depende do ponto P. Este número se denomina Potência do ponto.

Potência de Ponto



$$POT \quad P_T = d^2 - r^2$$

Relações Métricas do Triângulo $\triangle$



$$(I) \quad ah = bc$$

$$(II) \quad b^2 = am \quad || \quad c^2 = an$$

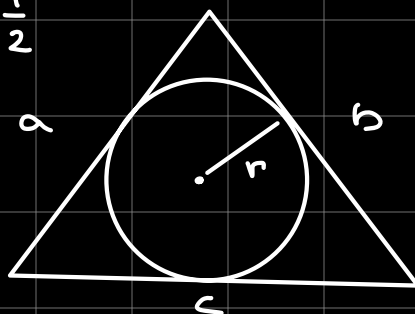
$$(III) \quad h^2 = mn$$

$$(IV) \quad \frac{1}{h^2} = \frac{1}{b^2} + \frac{1}{c^2}$$

Circunferência Inscrita

$$\Rightarrow p = \frac{(a+b+c)}{2}$$

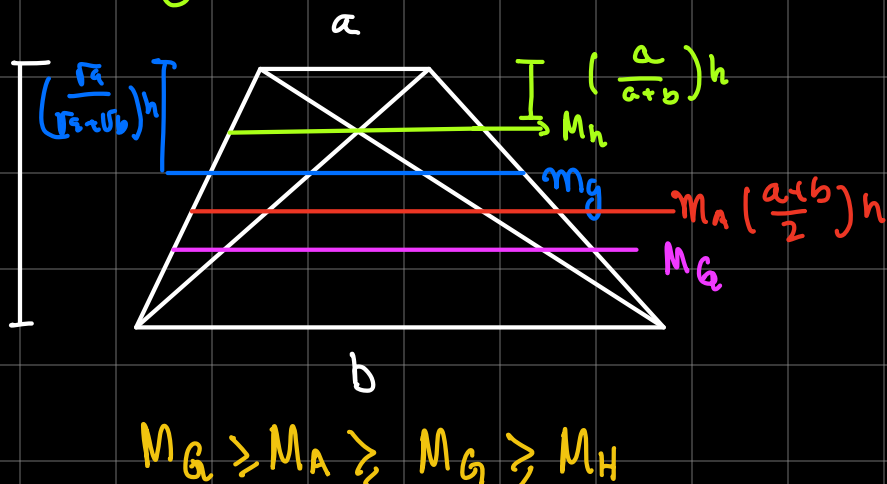
$$r = \frac{[ABC]}{p}$$



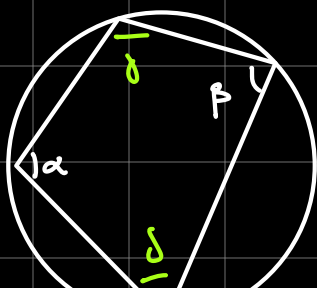
Quadriláteros Notáveis

• Paralelogramo

• trapézio



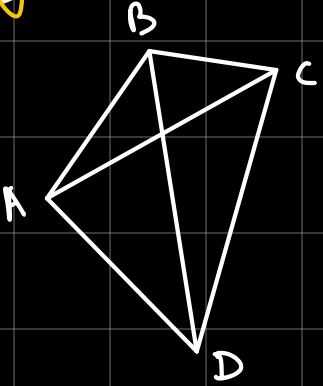
Quadriláteros Inscritíveis



1: É Inscriível se Ângulos opostos somam $\hat{=}$ 180 $\alpha + \beta = 180^\circ$

2: Ângulo reto: $\alpha = \beta$



3:  & $\overline{AC} \cdot \overline{BD} = \overline{AB} \cdot \overline{CD} + \overline{BC} \cdot \overline{AD}$

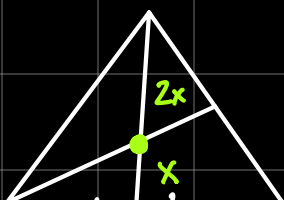
Quadriláteros Circunscritíveis

PITOT: é circunscritível se $AB + CD = AD + BC$
a soma de lados opostos é igual

Pontos Notáveis no Triângulo

① Baricentro (G)

Ponto de encontro das medianas

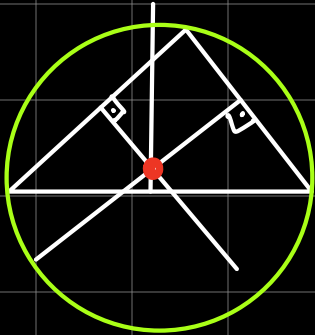


Centro de Gravidade

② Circuncentro (O)

→ Ponto de encontro das Mediatrizes

→ Centro da Circ. Circunscrita



③ Incentro